1. Squared Hasse predicate bridge: technical review

27 October 2026 UTC · T-504 · producer `f1c2dca`

The Lean theorem connects the Rust-style natural-number checks to uniqueness: both unsigned deviations from `p+1` square to at most `4p`, and `r²>16p`. Actual curve-cardinality facts remain outside the theorem.

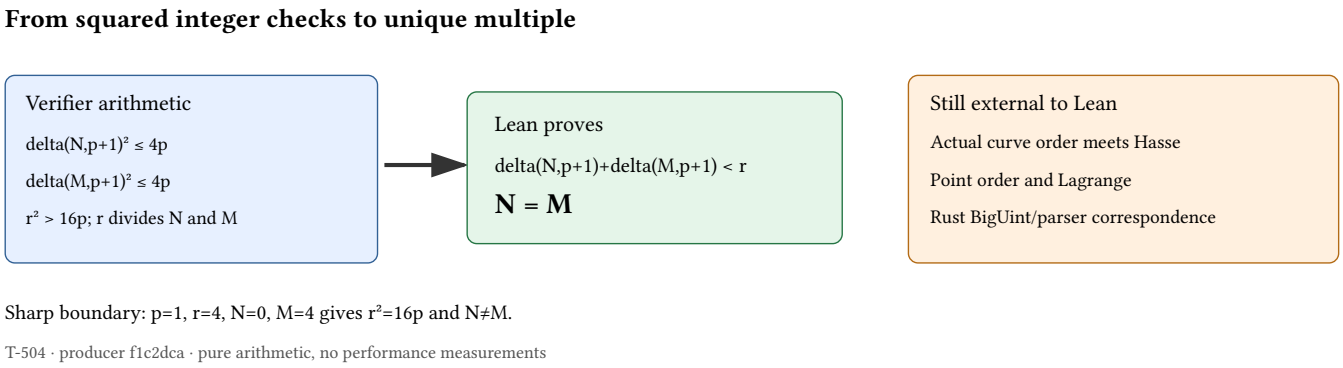

Figure 1: Checked arithmetic implication and separate curve-order obligations. No performance data were generated.

1.1. Separate verdicts

Claim	Machine	Fidelity	Overall
Nat squared-predicate uniqueness	passed	faithful scoped	validated scoped
Full P-192/P-224 order proof	missing bridges	partial	not verified

1.2. Evidence and controls

Producer source SHA-256: `26e3e7bbf28aad0f60f72137fca027427b435f05bcef9f5de5881d7979307c42`. Pinned Lean: 4.34.1. The exact-source build passed. The transitive audit found five new-module theorems, zero failures, zero `native_decide` exceptions, and only `propext`, `Quot.sound` or no axioms. The positive `p=101,r=41,n=m=82` application and sharp `p=1,r=4,n=0,m=4` boundary built and audited. The `sorry` false theorem compiled but its audit rejected `sorryAx`; the missing-module audit failed closed. All nine exits matched expectations and input hashes stayed unchanged. Commands, exits and hashes are in `machine-receipt.json` and raw logs.

Blind reviewer A derived the implication and sharp boundary without producer source. Reviewer B checked the Lean statements against the proof map and Rust predicate, disclosing extra generic process and neighboring Rust reads beyond the plan's literal list, but no sibling review access. These attestations were self-reported.

At the reviewed producer revision `f1c2dca`, the README omitted this new partial bridge. Documentation commit `03ca9ea` added the link and clarified the remaining obligations without changing the reviewed Lean source or machine inputs.

1.3. Remaining work

A complete certificate still needs a concrete prime field and elliptic group, exact point order and Lagrange divisibility, Hasse for the actual cardinality, Rust BigUint and parser correspondence, and P-192/P-224 input binding. The installed Lean kernel is the machine trust boundary; independent `lean4checker --fresh` was unavailable. No measured performance quantity changed, so existing performance graphs and dashboard values remain as they were.